Uncertain Giver Grasp in Bimanual Handover — V4

Where a fixed grasp transform fails, and what fixes it · Isaac Lab 3.0 / Isaac Sim 6.2 (PhysX) · CSCI 645, Fall 2026 · Yan Wen

- Two Franka arms with parallel-jaw grippers hand over a rigid bar; success = receiver holds it alone for 2 s
- V4: the bar's position inside the giver's grasp is unknown (−7…+15 mm along the bar, ±15 mm vertical)
- Fixed-transform reference (V3) collapses: 5.5% at +15 mm
- 0.5 s calibration from the same noisy sensor restores it: 94.5%
- History / residual PPO: at most a few cases in distribution, worse outside the training support
- Release v4.0.0: 132,000 logical held-out episodes, every checkpoint and raw record

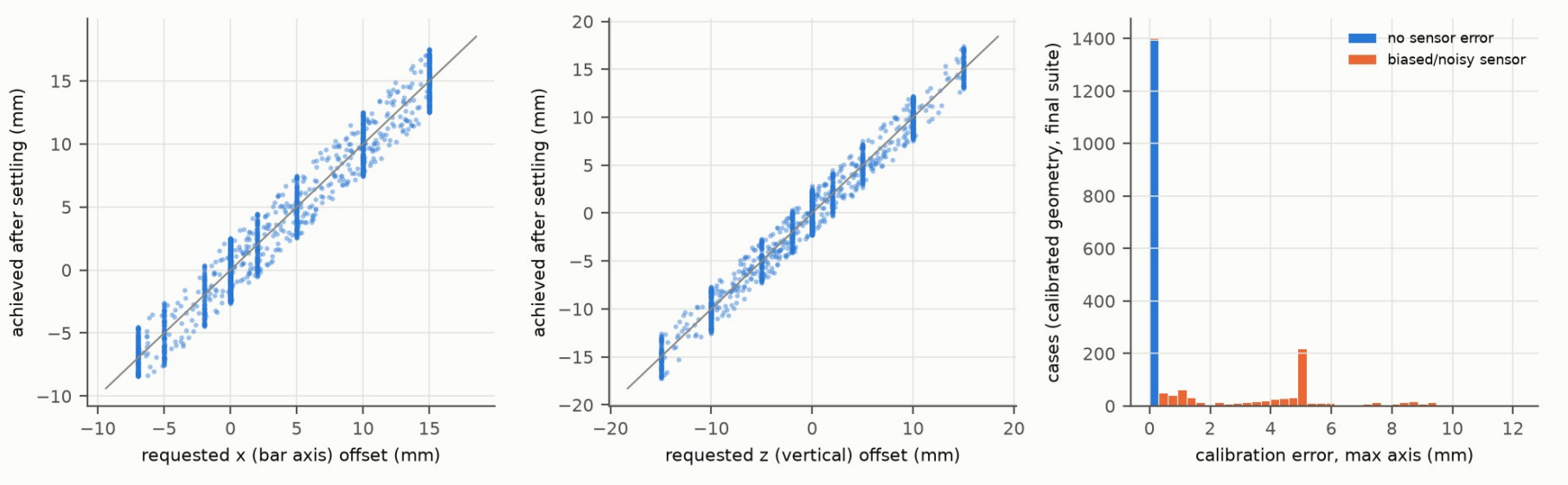
Why V4

earlier studies, quoted from their reports

- V1/V2: geometric coordination did most of the work; learned residuals ≈ small constant offsets
- V3: corrupted bar position (bias, noise, delay). Geometry + giver reference: 100% of 2,600 held-out cases
  - every learned policy also 100% → a ceiling; nothing could be ranked
- V3's key assumption: bar position = giver TCP + a known, rigid grasp transform
- V4 question: what if the bar sits somewhere else in the giver's fingers?
  - where does the fixed transform fail? do calibration, filtering, history or residual PPO help there?

The unknown in-grasp offset is physical, not a sensor trick

- Reset: bar pinned at nominal + joint displacement + in-grasp offset; giver closes at its unshifted pose
- Pin removed → only finger contact holds the bar; validity checked after settling; achieved offset recorded
- No weld, no teleport during the scored episode; the true offset is never visible to a controller
- Feasibility probe (development): y re-centred by the fingers → not a supported axis
  - toward the palm: −10 mm touches the giver hand (invalid), −7 mm valid; toward receiver: +15 mm stable
- Training support: x ∈ [−5, +15] mm, z ∈ [−10, +10] mm, 25% zero; evaluation adds −7 mm and ±15 mm (z)



Same information for every method

controller	how it locates the bar	fails when
fixed transform (V3)	giver TCP + nominal grasp vector	in-grasp offset (undetectable)
calibrated (0.5 s)	fixed transform + mean(measurement − fixed) over 10 frames	sensor bias (inherited)
EMA geometry (α 0.2)	filtered measurement only	bias, noise lag
learned residual (PPO, ±2 cm)	added to the calibrated parent or to nominal	outside training support

- Inputs for all: corrupted translation, proprioception, measured orientation, finger contacts, phase
- No controller reads the true offset, clean pose, case label or a future measurement (checked: A6/A8)
- Calibration window lies inside the common schedule (receiver pre-grasp at 1.0 s); timings include it

Develop → accept → pilot → freeze → train → validate → test

- Development: two registered baseline grids (≤ 36 configs) on 288 cases → calibrated parent (α 0.2, 0.5 s, 2.5 cm)
- Acceptance: V3 reproduced exactly (65/65 + 65/65), no hidden truth path, resets independent, resume works
- Pilots: 23,658,496 transitions (cap 24M) → V3 PPO settings kept; tag experiment-v4 frozen
- 25 formal runs: 5 profiles × seeds 111/222/333/444/555 × 10,027,008 transitions
- Checkpoint selection on 240 validation cases (2/4/6/8/10M; step 0 never eligible)
- Final: 12 conditions × 200 paired cases; 54 executed entries, 129,600 episodes
- Primary (mixed_id, selected): history − robust FF; history − best constant geometry; 10,000-draw paired bootstrap

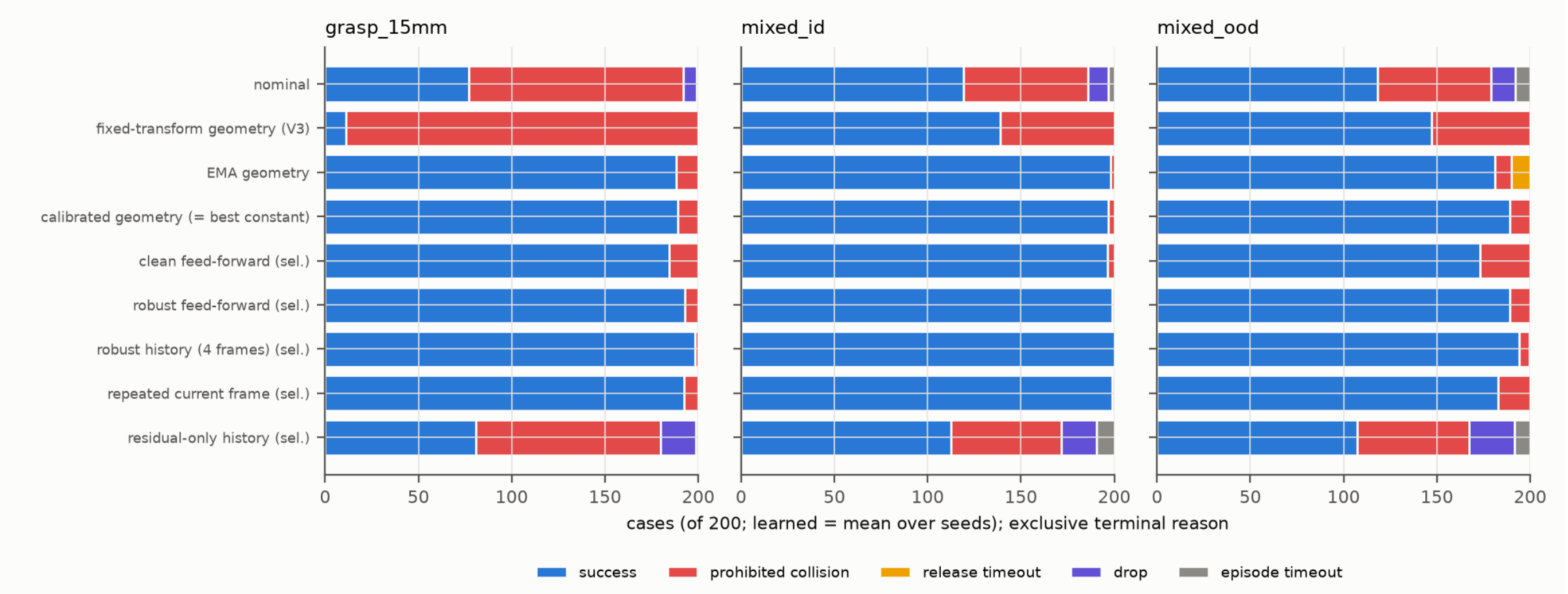
Result 1: the fixed transform breaks beyond ~+5 mm

held-out, 200 cases per condition

condition	nominal	fixed transform	EMA	calibrated
clean	100.0	100.0	100.0	100.0
grasp_5mm	64.0	95.5	100.0	100.0
grasp_10mm	47.0	39.5	100.0	100.0
grasp_15mm	38.5	5.5	94.0	94.5
bias_5mm	72.5	100.0	100.0	100.0
mixed_id	59.5	69.5	99.0	98.5
mixed_ood	59.0	73.5	90.5	94.5
boundary_x_minus_7	48.0	96.0	96.0	96.5

- Calibrated − fixed on grasp_15mm: +89.0 [+84.5, +93.0] pp
- Failures are collisions with the bar end: the target lies inside the bar
- Vertical and palm-side offsets are tolerated by the fixed transform
- Sensor-only corruption does not hurt it (it ignores the sensor while the giver holds)

Failure modes by method



Result 2: learned history adds at most a few cases

mixed_id, selected checkpoints, paired bootstrap

contrast	difference [pp] (95% CI)	97.5% Bonferroni
history − robust feed-forward	+0.6 [+0.0, +2.0]	[+0.0, +2.1]
history − best constant geometry	+1.5 [+0.0, +3.5]	[+0.0, +3.5]

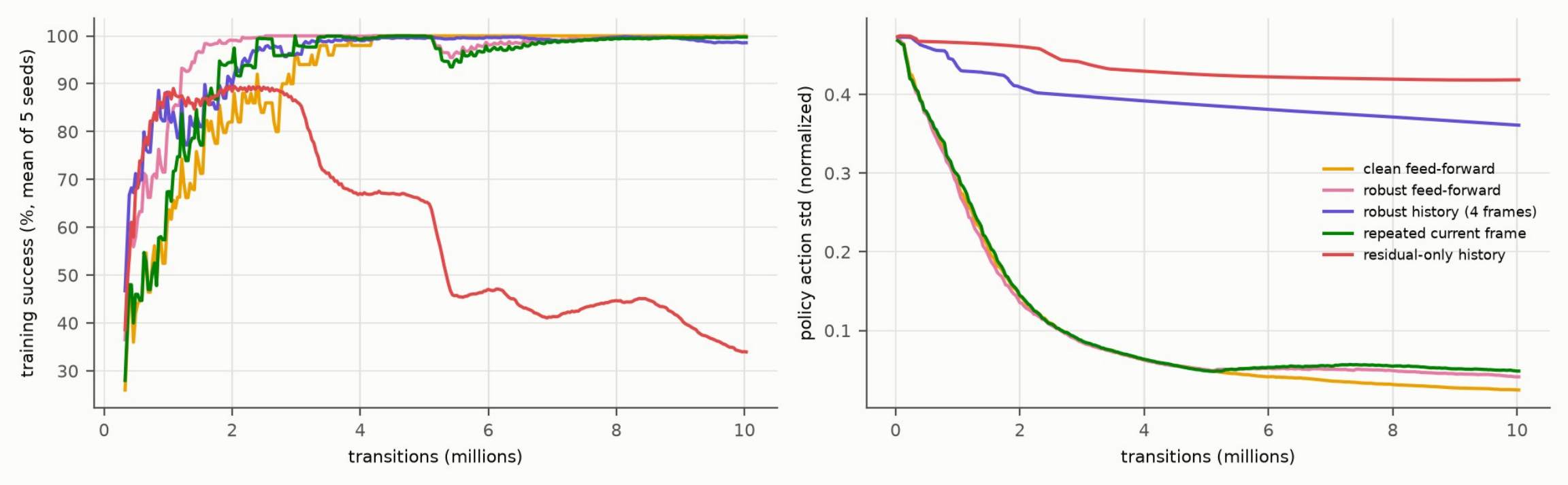
- Robust history: 100.0% on mixed_id; calibrated baseline 98.5%
- Both intervals touch zero → no improvement established; no equivalence claimed
- Deterministic baseline is one row, never counted as five seeds
- Final (10M) checkpoints instead: history − best constant = −1.5 [−5.5, +1.7] pp

Result 3: fragile outside the training support



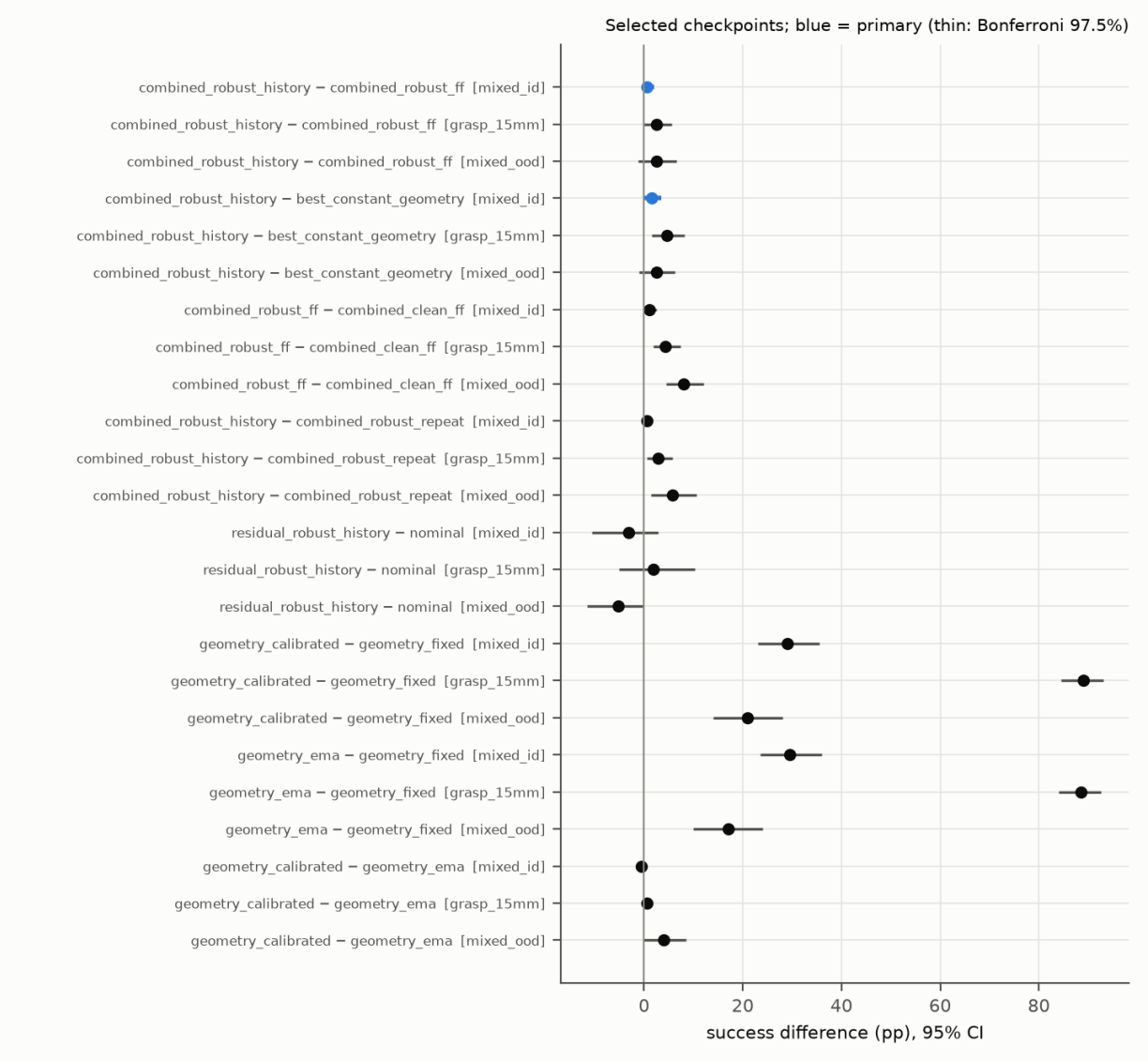
- Palm-side −7 mm (outside [−5, +15]): history 65.2% vs calibrated parent 96.5%
  - per seed 59.0 / 61.5 / 85.5 / 72.5 / 47.5%
- Over all non-clean conditions: history − parent = −2.3 [−3.9, −0.7] pp
- Residual-only on nominal: 56.3% vs nominal 59.5% (mixed_id)

Result 4: late training instability (again)



- History final checkpoints: bias_5mm per seed 100.0 / 100.0 / 60.5 / 84.5 / 100.0%
- Residual-only collapses after ~3M transitions: final 33.3% on mixed_id
- Selected steps ({step: runs}): 2031616: 4, 4030464: 2, 6029312: 6, 8028160: 3, 10027008: 10
- Validation-based selection avoided most of the late damage

Paired effects (selected checkpoints)



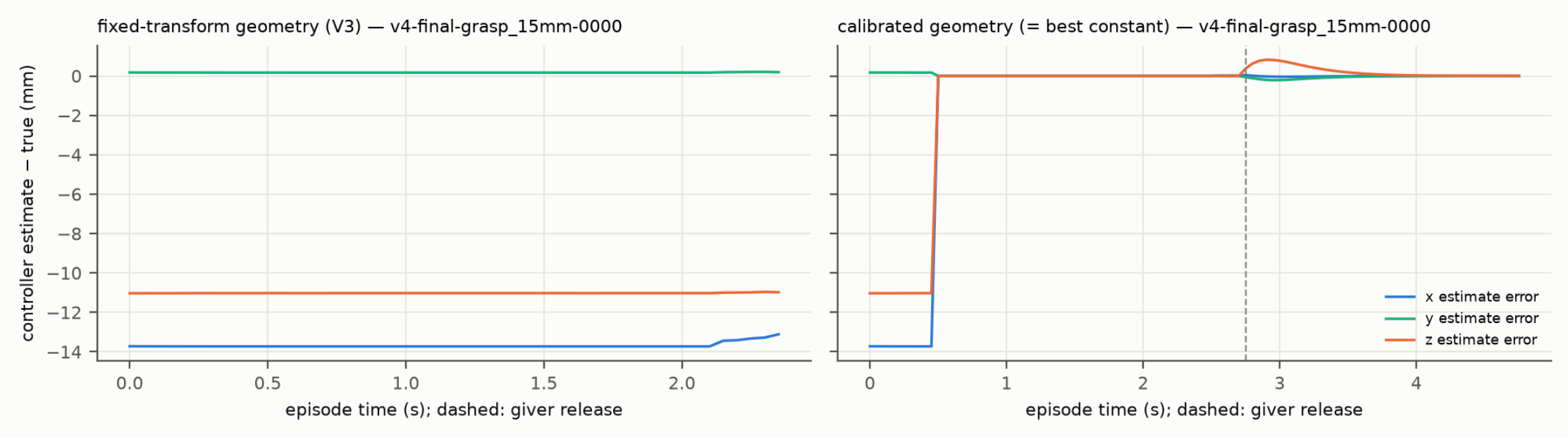
What do the learned residuals do?

60-case subset (5 per condition), selected history runs

run	policy	reversed	current only	dev constant	parent
history s111	57	57	55	60	60
history s222	58	57	57	60	60
history s333	59	60	59	60	60
history s444	59	60	58	60	60
history s555	55	51	51	59	60

- A constant fitted on development records does as well as each policy (or better)
- Reversing or removing frame order changes only a few outcomes → little temporal use
- Residual-only seeds 222/333 learned ~1 cm offsets (+9.1, −11.0, −11.8 mm) and fail
- Untrained step-0 checkpoints reproduce their parents (60/60)

Repeatability, traces and videos



- Repeatability: 2,160 of 2,160 reruns (batch 6/20 × 3 process seeds) reproduce the outcome
- Six-case trace runs agree with formal records in 323/324 pairs
- 10 clips + 53 s montage by a frozen rule; replays = evaluations
- No selected history policy failed a mixed_id case → no failure clip, no substitute
- Trace: fixed transform keeps the offset as a constant error until collision; calibration removes it at 0.5 s

Answer to the V4 question

- The fixed-transform grasp reference breaks beyond ~+5 mm toward the receiver (39.5% at +10 mm)
- A 0.5 s calibration from the same noisy measurements is the effective fix (+29.0 [+23.0, +35.5] pp on mixed_id)
- Calibration inherits sensor bias; EMA filtering is a close second in distribution
- Residual PPO and short history add at most a few in-distribution cases (primary CIs touch zero)
- …and they are fragile outside the training support and late in training
- Practical lesson: estimate what you can measure (calibrate) before learning a residual

Limitations

- Static in-grasp offsets only; dynamic slip not performed (no slip-robustness claim)
- Synthetic translation sensor, one bar, simulation only (no vision, no sim-to-real)
- Calibration assumes accurate proprioception/orientation and a still giver during 0.5 s
- Near-ceiling primary condition (calibrated 98.5%): small effects, no equivalence claim
- Training support widened once during development (logged); −7 mm boundary was outside it and outside validation
- 380 invalid resets at the −7 mm palm limit count as failures for every method
- Five seeds per profile

Everything is released

github.com/ywen0511/CSCI645-Project · release v4.0.0

- Tags: experiment-v4 (simulator source of every run) and v4.0.0 (analysis, report, packaging)
- Every checkpoint (formal, pilot, smoke, resume, killed attempt), raw record, log, ledger, video
- Manifest + per-archive member inventories + SHA256SUMS; checkpoint index with asset/member/SHA-256
- docs/v4/REPRODUCE.md: anonymous download → verify → regenerate tables/figures → replay/reload/resume
- Compute: 16.0 A100 GPU-hours for the whole continuation (cap 100)
- AI assistance: implemented, run and written by an AI coding agent (Claude via Claude Code) for the owner